

Mobility Lens — Assignment Report

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Package name: edu.bu.saichava.mobilitylens

Emulator used: Pixel 10 Pro, API 37.1

SDK Configuration

Setting	minSdk	targetSdk	compileSdk
Value	26 (Android 8.0, Oreo)	37	37

The minSdk defines the oldest Android version the app can install on. API 26 was chosen because it still covers roughly 98% of active devices while giving full support for modern Jetpack Compose APIs, with no compatibility workarounds needed. The targetSdk tells the system which Android version the app was designed and tested against, so the OS applies that version's behavior instead of falling back to older defaults. The compileSdk sets which version of the Android SDK libraries the code compiles against, giving access to the newest APIs and lint checks, and is generally kept at or above targetSdk.

Key Files and Their Purpose

MainActivity.kt (app/src/main/java/edu/bu/saichava/mobilitylens/MainActivity.kt) is the app's entry point. Its onCreate() sets the Compose content, wrapping the UI in the custom theme and a Scaffold.

AndroidManifest.xml (app/src/main/AndroidManifest.xml) declares the app's components, including MainActivity as the launcher activity, plus the app's icon and label. This app requests no permissions, since it uses no sensors, network, or storage.

build.gradle.kts (Module :app) configures how the app module builds: namespace, applicationId, the minSdk/targetSdk/compileSdk values, the Compose build feature, and the library dependencies.

The version catalog (gradle/libs.versions.toml) centralizes dependency names and versions in one file, so build.gradle.kts can reference them (e.g., libs.androidx.compose.ui) instead of hardcoding version strings.

strings.xml (app/src/main/res/values/strings.xml) stores all user-facing text as named string resources, referenced in Kotlin via R.string.*. This project defines 26 entries covering the introduction, six mobility dimensions, the navigation buttons, the input label, and validation feedback.

Rotation Observation

To test this, I navigated to a dimension other than the first, typed text into the app-name field, tapped Check to display a feedback message, and then rotated the emulator.

After rotating, the screen reset completely: the selected dimension returned to the first one, the typed text disappeared, and the feedback message was gone. None of the three pieces of UI state survived the rotation.

This happens because rotating counts as a configuration change, and by default Android destroys and recreates the Activity when one occurs, re-running onCreate() as if the app had just launched. currentIndex is declared with remember { mutableIntStateOf(0) }, while appName and feedbackMessage use remember { mutableStateOf(...) }. These state holders survive recomposition within the same Activity instance, but they do not survive Activity destruction and recreation.

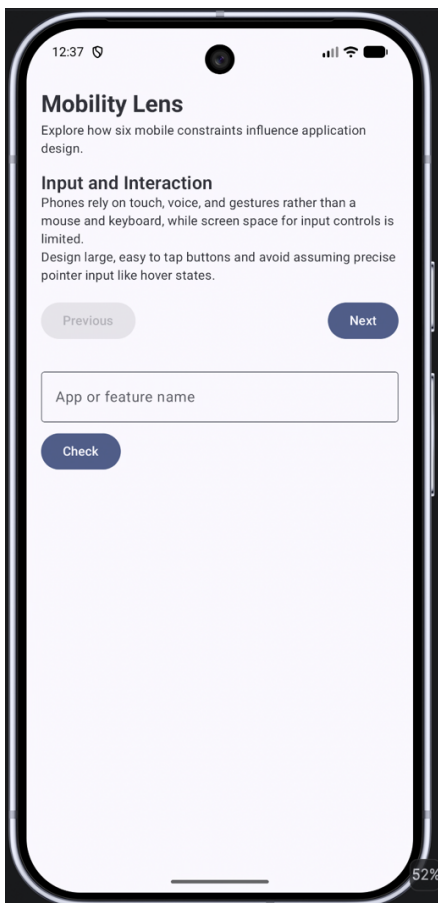
Preserving this would require rememberSaveable instead, which writes values into the saved-instance-state bundle handed back to the recreated Activity. This matters in real apps because a user who rotates their phone, or whose app is briefly killed by the system to free memory, should not lose in-progress input or their place in a multi-step flow.

Mobile Design Decision

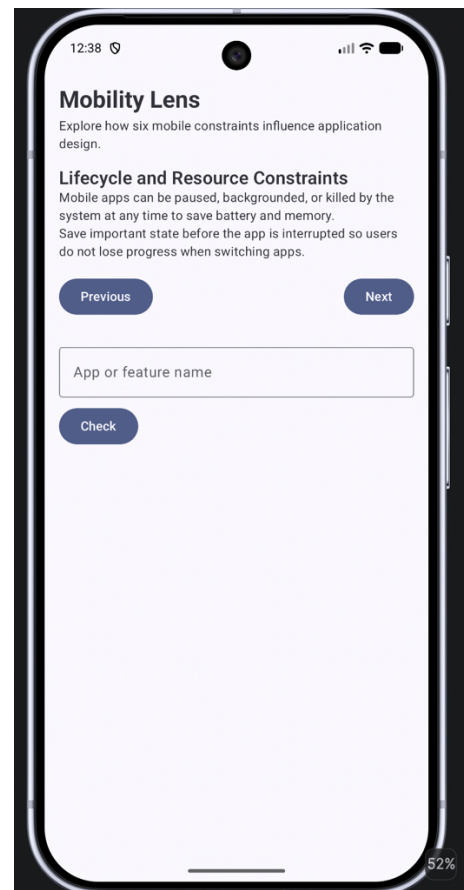
One deliberate mobile-appropriate decision was customizing the app's typography rather than leaving the default Material styles untouched. The headlineMedium and titleLarge styles were given a bolder weight, a larger size, and slightly wider letter spacing, so the app title and the current dimension name clearly stand out from the body text on a small screen. Because phone screens are small and users scan content quickly, a clear visual hierarchy with only a few text sizes lets users immediately identify which dimension they are viewing, which matters far less on a desktop application with a large screen and room for supplementary navigation like sidebars or breadcrumbs.

Screenshots

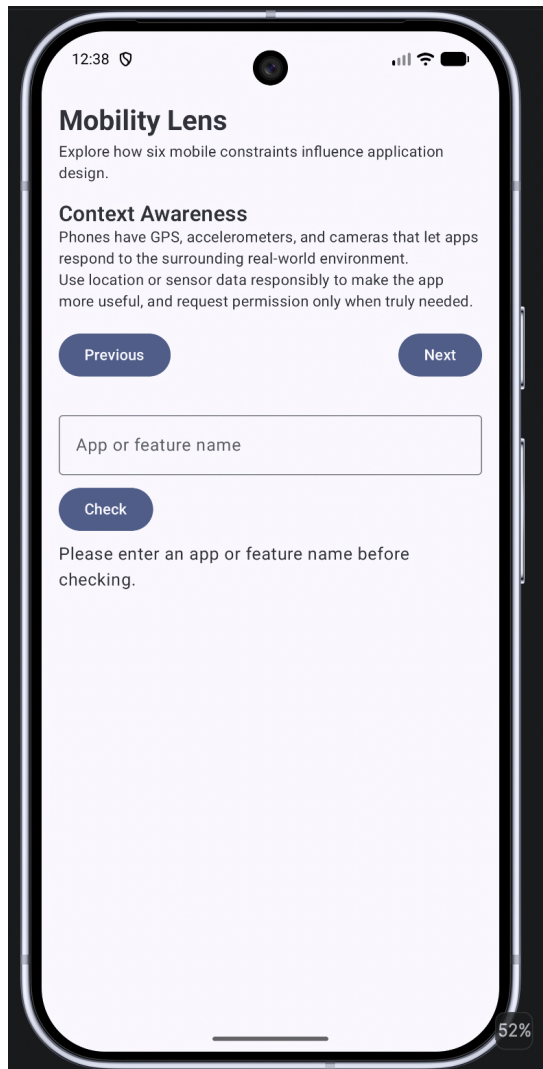
1. Running application on the first dimension



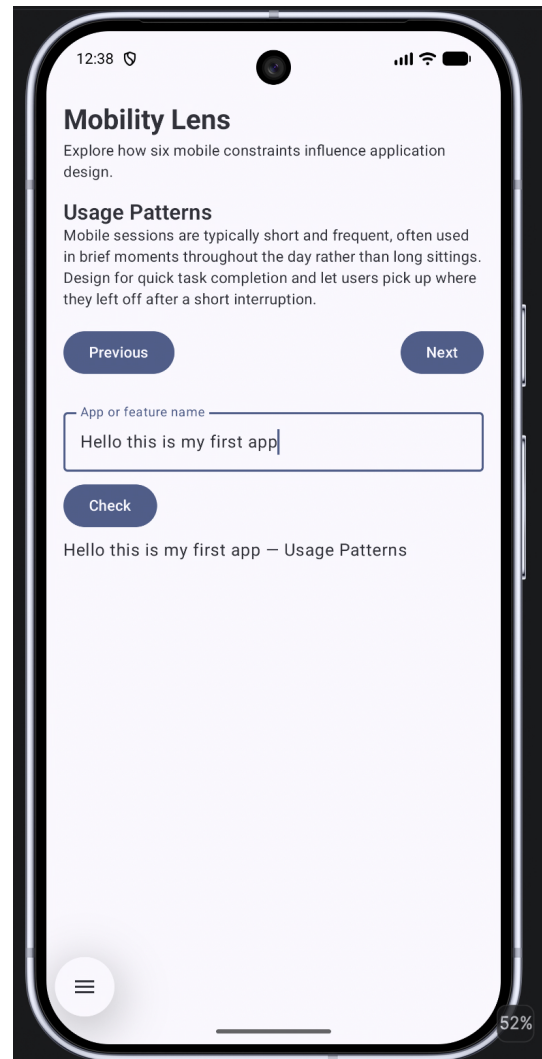
2. A different dimension after user interaction



3. Validation response to blank input



4. Successful response to valid input



Collaboration Disclosure

I did not collaborate with any classmates on this assignment.

Generative AI Disclosure

Tool or model used: Claude, an AI assistant developed by Anthropic, accessed through the claude.ai web interface.

Prompts: Prompts I used included:

- Explaining the code for Previous/Next buttons that update the currently shown dimension using Compose state, with the buttons disabled at the first and last dimension.
- Explaining the code for an OutlinedTextField and a Check button that validates blank input and combines the entered text with the current dimension name.

- Diagnosing why a build error occurred in strings.xml and how to fix it.
- Revising, and Grammar Correction for the written report and this README.

Relevant suggestions it produced: Claude explained core Compose concepts such as remember, mutableStateOf, Modifier; the Previous/Next navigation row, the OutlinedTextField, and the validation button; produced a customized Type.kt with bolder headline/title text styles; and identified that a build error was caused by an unescaped apostrophe in a strings.xml entry.

What I accepted, changed, or rejected: I accepted the overall Compose state pattern (remember/mutableStateOf). I wrote the six dimension descriptions and developer-implication text in strings.xml entirely in my own words rather than using Claude. I changed one suggested variable declaration from var to val after Claude explained why val was more appropriate there. I accepted the apostrophe/build-error fix and the unused-import cleanup after understanding why each was needed.

How I verified the resulting code: After each change, I ran the app on the Pixel 10 Pro emulator in Android Studio and manually tested the specific behavior involved: tapping Next/Previous through all six dimensions and confirming the buttons disabled at each end, typing into the text field and tapping Check with both blank and valid input, and rotating the emulator to observe and confirm the state-loss behavior described in the Rotation Observation section. I read through each generated file myself before accepting it. I confirmed that the project built successfully with no errors. The remaining lint notices only identify newer dependency versions.